Economic Complexity - Research Proposal

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Attitudes toward climate change vary across regions, and social connections may be one factor shaping these differences. Social ties provide channels through which information, opinions, and norms can spread, so strongly connected regions may be exposed to more similar discussions and viewpoints (Goldberg et al., 2019; Hannibal & Vedlitz, 2018).

Previous network studies suggest that social connectedness is associated with similarity in political attitudes and behaviour. Using Facebook connections between European regions, Bailey et al. (2020) show that social connectedness is related to political attitudes. Similarly, Basak et al. (2024) find that more strongly connected U.S. counties exhibit more similar presidential voting patterns, while Firsin (2026) shows that immigration attitudes can spread through social networks.

Climate-specific research points to a similar mechanism. Climate discussion has been linked to later climate concern (Goldberg et al., 2019), as well as to perceived climate knowledge and policy preferences (Hannibal & Vedlitz, 2018). This suggests that social ties may also be associated with similarity in climate attitudes by facilitating the exchange of climate-related information and norms.

Existing research links social connectedness to attitudes on different topics, but whether socially connected regions actually have more similar climate attitudes remains unknown. Therefore, our research question is whether pairs of European NUTS3 regions that are more socially connected are more similar in their attitudes toward climate change. This question is relevant because climate policy depends on public concern, perceived responsibility, and understanding of policy. If climate attitudes are more similar across socially connected regions, this would suggest that the social structure connecting regions may be important for understanding how climate-related views are distributed. An important limitation of the research is that available observational network data cannot separate social influence from homophily or shared contextual factors (Shalizi & Thomas, 2011).

Our empirical approach combines two main data sources: the Facebook Social Connectedness Index (2026), capturing the relative intensity of online social ties across regions, and the ESS CRONOS-3 Wave 6 survey (2025), which provides individual-level perspectives on environmental issues across 11 European nations. In this network structure, European NUTS3 territories serve as nodes, connected by edges weighted according to their mutual social tie strength. Using the ESS data, we construct aggregated regional means for key metrics: degree of environmental worry, sense of individual obligation to mitigate climate impacts, and self-reported awareness of domestic environmental strategies. The analysis evaluates dyadic pairs of regions. For every pair, we compute dissimilarity in mean environmental perspectives and estimate its association with social connection strength.

While both data sources are accessible, an important constraint can be sample sizes. Should NUTS3 estimates prove insufficient, the network metrics will be scaled to NUTS2 units or alternative spatial aggregations.

We anticipate that region pairs featuring stronger connections display smaller differences in climate perspectives, even after controlling for physical proximity and national background. This alignment is expected to be more pronounced regarding public concern and individual responsibility, which are frequently reinforced through interpersonal interactions and social norms.

REFERENCES

Bailey, M., Kuchler, T., Russel, D., State, B., & Stroebel, J. (2020). Social connectedness in Europe. SocArXiv. https://doi.org/10.31235/osf.io/3wh67

Basak, E., Tafti, A., & Pang, M.-S. (2024). The role of online and geographically distant social networks in political decision-making: Empirical evidence from the 2016 United States presidential election. Journal of the Association for Information Systems, 25(3). https://doi.org/10.17705/1jais.00851

Firsin, O. (2026). Social networks and spread of immigration attitudes. Economic Inquiry. https://doi.org/10.1111/ecin.70042

Goldberg, M. H., van der Linden, S., Maibach, E., & Leiserowitz, A. (2019). Discussing global warming leads to greater acceptance of climate science. Proceedings of the National Academy of Sciences of the United States of America, 116(30), 14804–14805. https://doi.org/10.1073/pnas.1906589116

Hannibal, B., & Vedlitz, A. (2018). Social capital, knowledge, and the environment: The effect of interpersonal communication on climate change knowledge and policy preferences. Sociological Spectrum, 38(5), 277–293. https://doi.org/10.1080/02732173.2018.1502108

Shalizi, C. R., & Thomas, A. C. (2011). Homophily and contagion are generically confounded in observational social network studies. Sociological Methods & Research, 40(2), 211–239. https://doi.org/10.1177/0049124111404820